

Homework 9

Due: Wednesday, 31st July 2024 at 1:00 PM

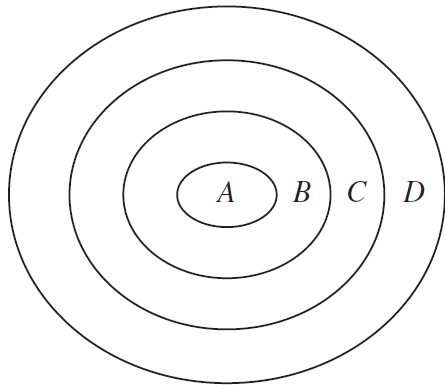
Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

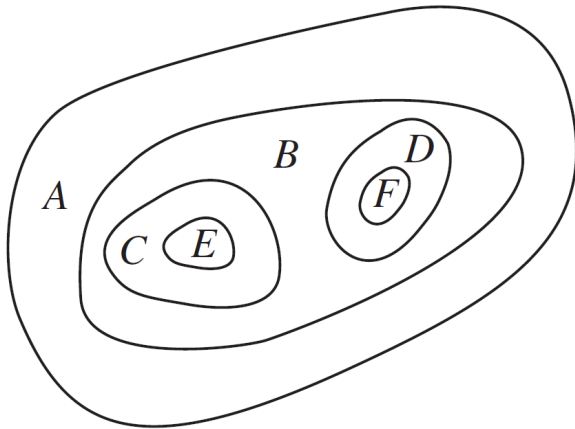
Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

In Questions 1–2 construct the dual graph for the map shown. Then find the number of colors needed to color the map so that no two adjacent regions have the same color.

Question 1.

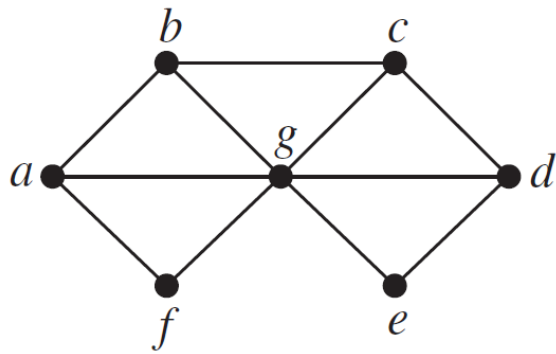


Question 2.

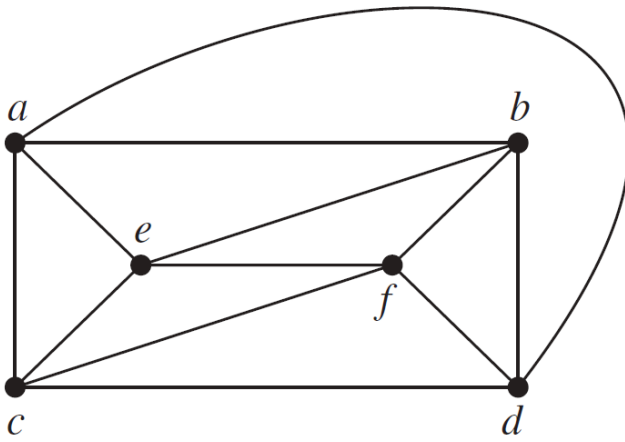


In Questions 3–5 find the chromatic number of the given graph.

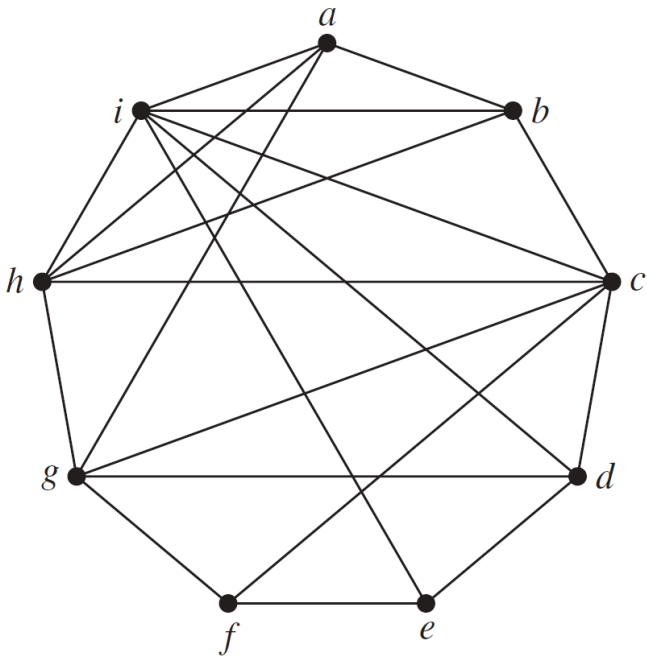
Question 3.



Question 4.



Question 5.



Question 6.

Which graphs have a chromatic number of 1?

Question 7.

What is the least number of colors needed to color a map of the United States? Do not consider adjacent states that meet only at a corner. Suppose that Michigan is one region. Consider the vertices representing Alaska and Hawaii as isolated vertices. Explain your reasoning.

Question 8.

Show that a simple graph that has a circuit with an odd number of vertices in it cannot be colored using two colors.

Question 9.

How many different channels are needed for six stations located at the distances shown in the table, if two stations cannot use the same channel when they are within 150 miles of each other?

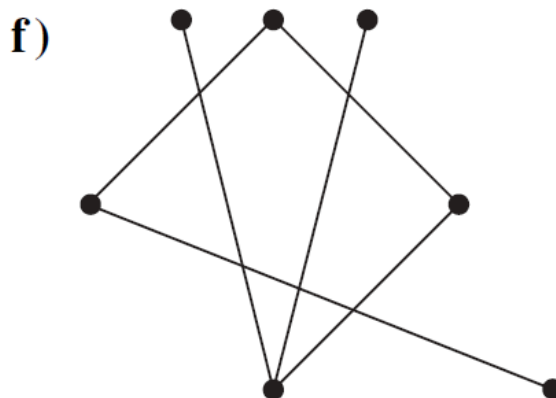
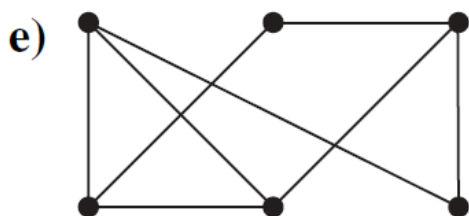
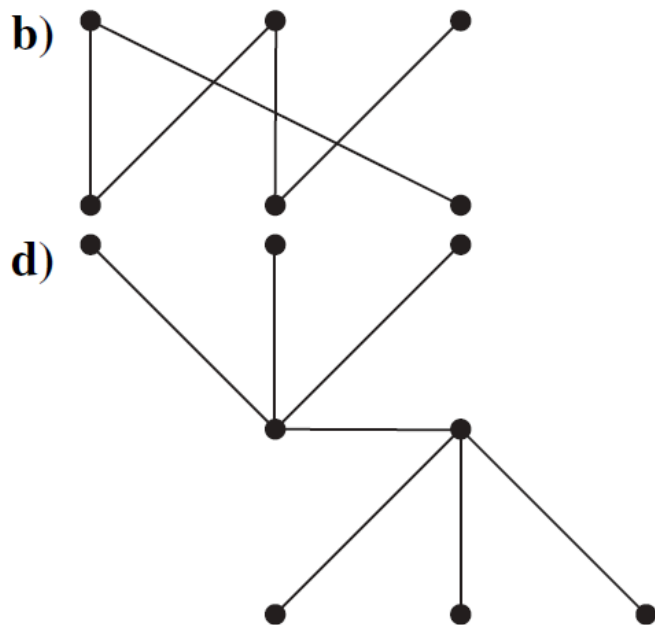
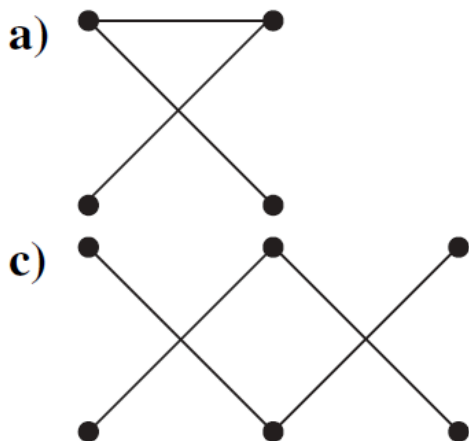
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>
<i>1</i>	—	85	175	200	50	100
<i>2</i>	85	—	125	175	100	160
<i>3</i>	175	125	—	100	200	250
<i>4</i>	200	175	100	—	210	220
<i>5</i>	50	100	200	210	—	100
<i>6</i>	100	160	250	220	100	—

Question 10.

A zoo wants to set up natural habitats in which to exhibit its animals. Unfortunately, some animals will eat some of the others when given the opportunity. How can a graph model and a coloring be used to determine the number of different habitats needed and the placement of the animals in these habitats?

Question 11.

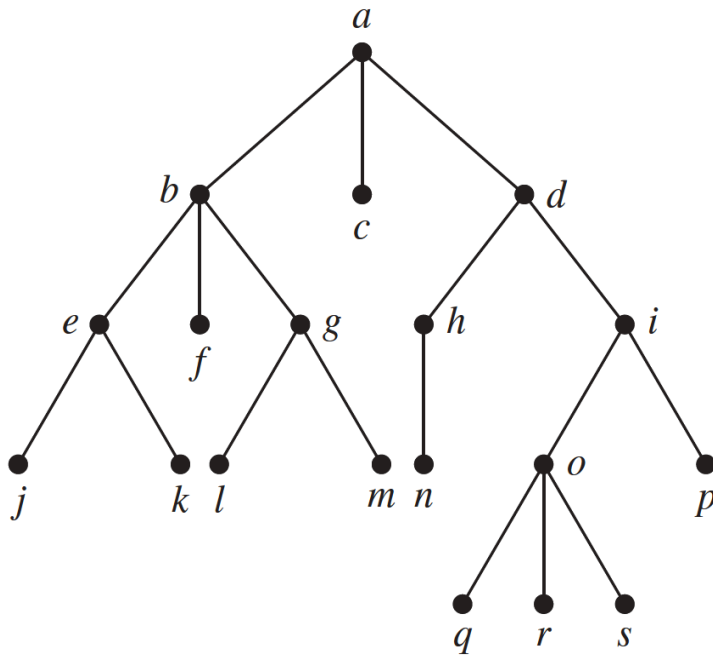
Which of these graphs are trees?



Question 12.

Answer these questions about the rooted tree illustrated.

- a) Which vertex is the root?
- b) Which vertices are internal?
- c) Which vertices are leaves?
- d) Which vertices are children of j ?
- e) Which vertex is the parent of h ?
- f) Which vertices are siblings of o ?
- g) Which vertices are ancestors of m ?
- h) Which vertices are descendants of b ?



Question 13.

Which complete bipartite graphs $K_{m,n}$, where m and n are positive integers, are trees?

Question 14.

How many vertices does a full 5-ary tree with 100 internal vertices have?

Question 15.

A **complete m-ary tree** is a full m-ary tree in which every leaf is at the same level. How many vertices and how many leaves does a complete m-ary tree of height h have?